Curriculum Vitae

Yue Ma — Experimental Nuclear Physicist

Senior Research Scientist (permanent), Few-body Systems Physics Laboratory, RIKEN | Born 1979-07-19 | y.ma@riken.jp | ymaphys.github.io | ORCID 0000-0001-8412-8308

1. Profile

Experimental nuclear physicist and Senior Research Scientist (permanent) at RIKEN, with more than 15 years of front-line research and leadership at major international facilities in Japan (J-PARC, SPring-8, KEK) and Germany (GSI, DESY), and a long record of training young researchers and building international collaborations. **Currently experiment spokesperson of** J-PARC E73 **(hypertriton lifetime), and proposal spokesperson of** SPring-8 LEPS2 **(nucleon short-range correlations) and the** J-PARC **antideuteron experiment.**

2. Academic Leadership & Appointments

- J-PARC E73, **hypertriton lifetime — Experiment spokesperson.** Beam time completed; the $^{4}_{\Lambda}$H lifetime and $^{3,4}_{\Lambda}$H production cross sections are published; the $^{3}_{\Lambda}$H lifetime analysis is ongoing.
- SPring-8 LEPS2, **nucleon short-range correlations — Proposal spokesperson.** Probing short-range correlations between nucleons with the Λ hyperon as a probe (ongoing).
- J-PARC **antideuteron experiment — Proposal spokesperson.** Antideuteron–nucleus interaction and annihilation mechanism (ongoing).

3. Signature Achievements

1. **Discovery of the $K^{-}pp$ ($\bar{K}NN$) kaonic nuclear bound state** — as a founding member, developed the multi-hit TDC and extracted the signal by polarization analysis, giving the first evidence for a deeply bound $K^{-}pp$ state (J-PARC E15). Phys. Lett. B **789** (2019); Phys. Rev. C **102** (2020).
2. **Precise lifetime measurement of the $^{4}_{\Lambda}$H hypernucleus** — as spokesperson and corresponding author, proposed the key idea of tagging the fast π^{0} with a single high-energy γ and led the measurement, $\tau = 206 \pm 8 \pm 12\,\mathrm{ps}$ (J-PARC E73). Phys. Lett. B **845** (2023); **873** (2026).
3. **First observation of a $\Lambda\pi$ structure near the $\bar{K}N(I{=}1)$ threshold** — led the Belle data analysis as first author. Phys. Rev. Lett. **130** (2023).
4. **First application of** TES **microcalorimeters to hadronic-atom spectroscopy** — implemented the SDD data analysis and energy calibration at sub-eV precision (J-PARC E62). Prog. Theor. Exp. Phys. **2016**, 091D01; Phys. Rev. Lett. **128** (2022).
5. **First demonstration of energy-resolved muon spin spectroscopy (μSR)** — as first and corresponding author, introduced event-by-event positron-energy selection with lead-glass Cherenkov calorimeters and led the demonstration, a $\sim$30% figure-of-merit gain at no cost in beam time (J-PARC MLF). Nucl. Instrum. Meth. A, in press (2026).

4. Education

2006–2009 **Doctor of Science**, Tohoku University (Sendai)

2004–2006 M.S. in Physical Science, Tohoku University (Sendai)

1998–2002 **B.S. in Nuclear Physics**, Lanzhou University

5. Appointments

2012–now **Senior Research Scientist (permanent)**, RIKEN

2009–2012 Postdoctoral Researcher, Helmholtz Institute Mainz, Germany

2002–2004 Research Trainee, Institute of Modern Physics, CAS

6. Grants and Fellowships

2026–2027 Grant-in-Aid for Transformative Research Areas (A), Publicly Offered Research: search for a Σ-hypertriton ($^3_\Sigma$H) bound state at J-PARC (within E. Hiyama's research area). **PI**

2024–2028 Grant-in-Aid for Scientific Research (A): ρ-meson mass modification in nuclei with photon beams. Co-investigator

2023–2025 RIKEN Incentive Research Project: exploring matter–antimatter interactions. **PI**

2017–2020 JSPS Grant-in-Aid for Young Scientists (A) 17H04842: solving the hypertriton lifetime puzzle. **PI**

7. Student Supervision

2017–2020 C. Han (PhD), Univ. of Chinese Academy of Sciences

2012–2014 Q. Zhang (PhD), Lanzhou University

2010–2011 D. Lin (PhD); Christina, Pascal (M.S.), Univ. of Mainz

8. Service & Invited Talks

2019–2021 Member, RIKEN Scientists' Assembly Steering Committee; coordinator / chief coordinator, RIKEN Incentive Research Fund (Physics)

2022 Invited talk, 14th Int. Conf. on Hypernuclear & Strange Particle Physics (HYP 2022, Prague)

2025 Invited talk, 15th HYP (HYP 2025, Tokyo)

9. Teaching & Languages

2018/23/24 Invited lecturer, Hadron Physics summer course, Univ. of Chinese Academy of Sciences

Languages Chinese (native) · English (fluent) · Japanese (professional working proficiency)

10. Selected Publications (by leading role and impact; authorship indicated)

More than **100 publications** (including Belle / Belle II collaboration papers, see the final section); the following are first-author / corresponding-author / spokesperson or otherwise leading contributions.

Note: hadron-physics papers are usually listed alphabetically by surname; the **first author** / **corresponding author** / **spokesperson** tags below mark the actual leading role.

1. **Y. Ma** *et al.*, *First observation of* $\Lambda\pi^+$ *and* $\Lambda\pi^-$ *signals near the* $\bar{K}N(I{=}1)$ *mass threshold in* $\Lambda_c^+ \to$

$\Lambda\pi^+\pi^+\pi^-$ *decay*, Phys. Rev. Lett. **130**, 151903 (2023) [Belle]. **[First author]**

2. T. Akaishi *et al.*, *Precise lifetime measurement of* ${}^4_\Lambda$*H hypernucleus using a novel production method*, Phys. Lett. B **845**, 138128 (2023). **[Spokesperson & corresponding author]**
3. T. Akaishi *et al.*, *Measurement of* ^{3,4}*He*(K^-,π^0) ${}^{3,4}_\Lambda$*H reaction cross section and evaluation of hypertriton* Λ *binding energy*, Phys. Lett. B **873**, 140163 (2026). **[Spokesperson]**
4. **Y. Ma** *et al.*, *Demonstration of energy-resolved muon spin spectroscopy using a Cherenkov calorimeter*, Nucl. Instrum. Meth. A, in press (2026). **[First author & corresponding author]**
5. **Y. Ma** *et al.*, γ*-ray spectroscopy study of* ${}^{11}_\Lambda$*B and* ${}^{12}_\Lambda$*C*, Eur. Phys. J. A **33**, 243 (2007). **[First author]**
6. S. Ajimura *et al.*, K^-pp*, a* $\bar{K}$*-meson nuclear bound state, observed in* ^{3}*He*$(K^-,\Lambda p)n$ *reactions*, Phys. Lett. B **789**, 620 (2019). **[Major author]**
7. T. Yamaga *et al.*, *Observation of a* $\bar{K}NN$ *bound state in the* ^{3}*He*$(K^-,\Lambda p)n$ *reaction*, Phys. Rev. C **102**, 044002 (2020). **[Major contributor]**
8. T. Hashimoto *et al.*, *Strong-interaction effects in kaonic-helium isotopes at sub-eV precision with X-ray microcalorimeters*, Phys. Rev. Lett. **128**, 112503 (2022). **[Major contributor]**
9. B. S. Henderson *et al.*, *Hard two-photon contribution to elastic lepton-proton scattering (OLYMPUS)*, Phys. Rev. Lett. **118**, 092501 (2017). **[Major contributor; led the PbF_2 luminosity monitor]**
10. K. Hosomi *et al.*, *Precise determination of* ${}^{12}_\Lambda$*C level structure by* γ*-ray spectroscopy*, Prog. Theor. Exp. Phys. **2015**, 081D01 (2015). **[Second author]**

11. Belle / Belle II Collaboration Papers

As a member of the Belle / Belle II collaboration since **2023**, co-author of **70+** collaboration papers; representative papers by theme:

I. B-meson CP violation & CKM matrix elements Time-dependent CP violation ($\sin 2\phi_1$, etc.), $|V_{cb}|/|V_{ub}|$, and the CKM angle ϕ_3.

1. *Measurement of* $\sin 2\phi_1$ *in* $B^0 \to J/\psi K^0_S$ *decays with a graph-neural-network flavor tagger*, Phys. Rev. D **110**, 012001 (2024) [Belle II].
2. *Measurement of CP violation in* $B^0 \to K^0_S\pi^0$ *decays*, Phys. Rev. Lett. **131**, 111803 (2023) [Belle II].
3. *Determination of* $|V_{cb}|$ *using* $\bar{B}^0 \to D^{*+}\ell^-\bar{\nu}_\ell$ *decays*, Phys. Rev. D **108**, 092013 (2023) [Belle II].
4. *Measurement of inclusive* $B \to X_u\ell\nu$ *partial branching fractions and* $|V_{ub}|$, Phys. Rev. D **113**, 032004 (2026) [Belle II].
5. *Determination of the CKM angle* ϕ_3 *from a combination of Belle and Belle II results*, J. High Energy Phys. **10** (2024) 143 [Belle/Belle II].

II. Rare decays & lepton-flavour universality First evidence for $B^+ \to K^+\nu\bar{\nu}$, $R(D^*)$ and other LFU tests, and τ lepton-flavour-violation searches.

1. *Evidence for* $B^+ \to K^+\nu\bar{\nu}$ *decays*, Phys. Rev. D **109**, 112006 (2024) [Belle II].
2. *Test of light-lepton universality with a measurement of* $R(D^*)$ *using hadronic* B *tagging*, Phys. Rev. D **110**, 072020 (2024) [Belle II].
3. *First measurement of* $R(X_{\tau/\ell})$ *as an inclusive test of the* $b \to c\tau\nu$ *anomaly*, Phys. Rev. Lett. **132**, 211804 (2024) [Belle II].
4. *Tests of light-lepton universality in angular asymmetries of* $B^0 \to D^{*-}\ell\nu$ *decays*, Phys. Rev. Lett. **131**, 181801 (2023) [Belle II].
5. *Measurement of the* $B^+ \to \tau^+\nu_\tau$ *branching fraction with a hadronic tagging method*, Phys. Rev. D **112**, 072002 (2025) [Belle II].
6. *First search for* $B \to X_s\nu\bar{\nu}$ *decays*, Phys. Rev. Lett. **136**, 231801 (2026) [Belle II].
7. *Search for lepton-flavor-violating* τ *decays to* $\ell\alpha$, J. High Energy Phys. **08** (2025) 155 [Belle].
8. *Search for a* $\tau^+\tau^-$ *resonance in* $e^+e^- \to \mu^+\mu^-\tau^+\tau^-$ *events*, Phys. Rev. Lett. **131**, 121802 (2023) [Belle II].

III. Charm/strange baryon spectroscopy & exotic hadrons(closely aligned with my own research) Ξ_c decays and CP, the first observation of $D_{s0}^*(2317)$ radiative decay, P_{cs} pentaquarks, and $\Xi^0 p/\Omega^- p/\Omega^- n$ dibaryon searches.

1. *Observation of the radiative decay* $D_{s0}^*(2317)^+ \to D_s^{*+}\gamma$, Phys. Rev. Lett. **136**, 241901 (2026) [Belle/Belle II].
2. *Search for* $P_{cs}(4459)$ *and* $P_{cs}(4338)$ *in* $\Upsilon(1S, 2S)$ *inclusive decays*, Phys. Rev. Lett. **135**, 041901 (2025) [Belle/Belle II].
3. *Search for CP violation in* $\Xi_c^+ \to \Sigma^+ h^+ h^-$ *and* $\Lambda_c^+ \to p h^+ h^-$ *decays*, Phys. Rev. D **113**, 032017 (2026) [Belle II].
4. *Observation of the singly Cabibbo-suppressed decays* $\Xi_c^+ \to pK_S^0$, $\Lambda\pi^+$, $\Sigma^0\pi^+$, J. High Energy Phys. **03** (2025) 061 [Belle/Belle II].
5. *Measurement of branching fractions of* $\Xi_c^0 \to \Xi^0\pi^0$, $\Xi^0\eta$, $\Xi^0\eta'$, J. High Energy Phys. **10** (2024) 045 [Belle/Belle II].
6. *Observation of* $B^+ \to \Sigma_c(2455)^{++}\bar{\Xi}_c^-$ *and* $B^0 \to \Sigma_c(2455)^0\bar{\Xi}_c^0$, Phys. Rev. D **112**, L051101 (2025) [Belle/Belle II].